1. Completed Tutorials and Learning Outcomes

During the first phase of the project, we completed a series of tutorials on Python, PyTorch, GitLab, Google Colab, LaTeX, and Scrum. These tutorials helped us understand core programming concepts, collaborative workflows, and the fundamentals of machine learning. Each team member practiced hands-on tasks, such as writing small Python scripts, training simple models, and managing repositories.

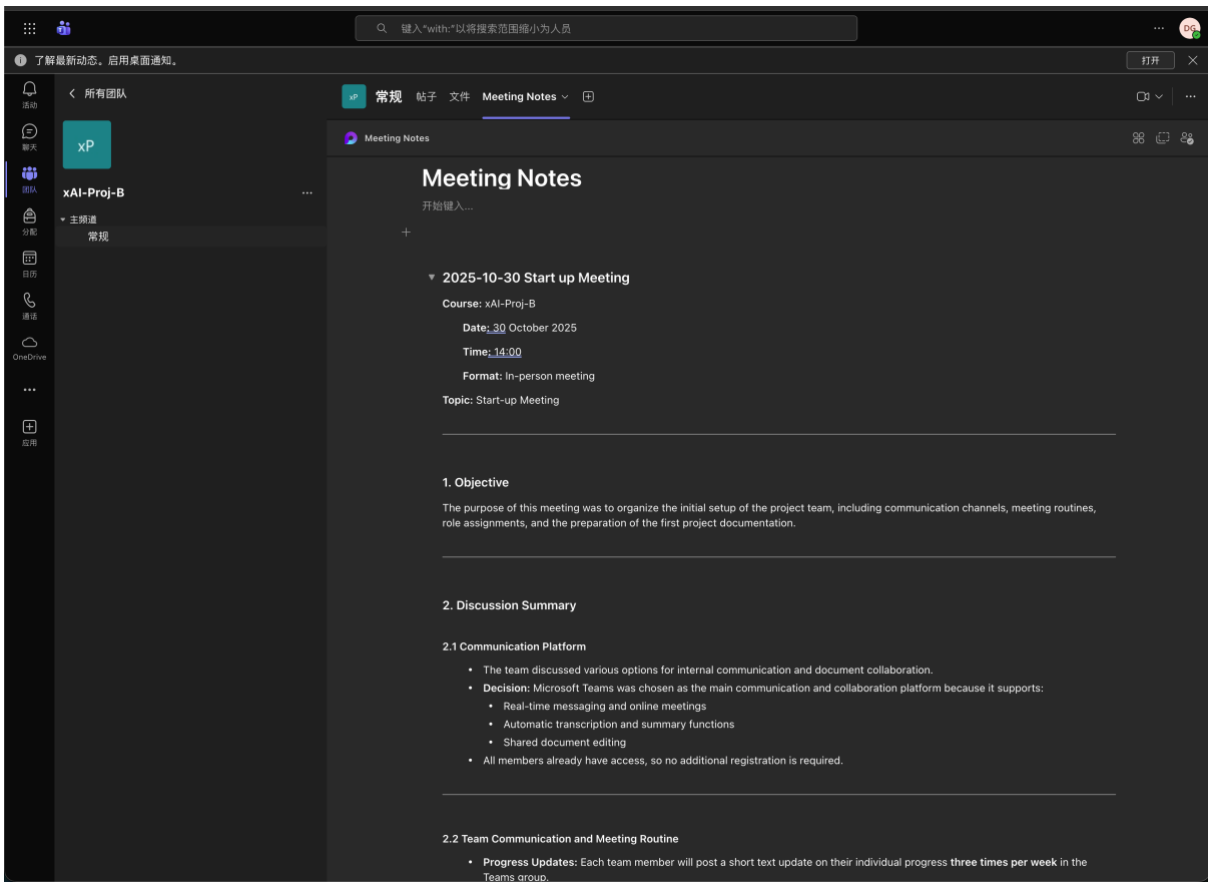
	Fatemeh Hajimohammadi	Dike Gan	Ruiqi Sun
Python fundamentals & scripting	Reviewed syntax and control flow; built small scripts for data loading and preprocessing.	Familiar with syntax, loops, functions, and classes. Proficient in NumPy, Pandas, and Matplotlib for data handling and visualization. Understands file I/O and basic data preprocessing.	Same as Dike Gan
PyTorch / timm / HuggingFace	Trained small CNNs; used timm pre-trained models and simple HuggingFace pipelines.	Understands ML/DL basics (datasets, overfitting, gradient descent). Can build and train models using PyTorch modules. Knows how to load pretrained models with timm and run inference with HuggingFace.	-
Version control (Git, GitLab, GitHub)	Followed a feature-branch + merge-request workflow; wrote clear commit messages and resolved conflicts.	Understands core Git commands (clone, commit, branch, merge); collaborates via GitHub pull requests and code reviews; able to resolve basic merge conflicts.	-
Google Colab for development	Ran GPU notebooks in Colab, mounted cloud drive and synced with GitHub; saved checkpoints for reproducibility.	Able to connect Google Drive, import datasets, and install dependencies. Runs training scripts using GPU/TPU environments in Colab. Knows how to save and export results from notebooks.	-
Reports in LaTeX	Produced structured reports in LaTeX (e.g. Overleaf), focusing	Familiar with LaTeX document structure, figures, and tables. Can write equations and format	-

	on clear tables, figures and citations.	academic reports using templates (e.g., IEEE). Understands reference and citation management.	
Scrum & collaboration	Held weekly planning and stand-ups; managed tasks with Trello and used a personal Trello to track learning goals.	Understands Scrum roles, sprints, and daily updates. Uses Trello for task management and sprint planning. Can reflect on progress during reviews and retrospectives.	-

2. Tool Setup and Configuration

After thorough discussion, we finalized the set of tools to be used in our project.

For team communication and collaboration, we chose **Microsoft Teams** as our main platform. We created a dedicated **Team Group** and set up a shared document called *Meeting Notes*. After each meeting, the **Communication Master** records the minutes and uploads them to this shared document for all members to review.

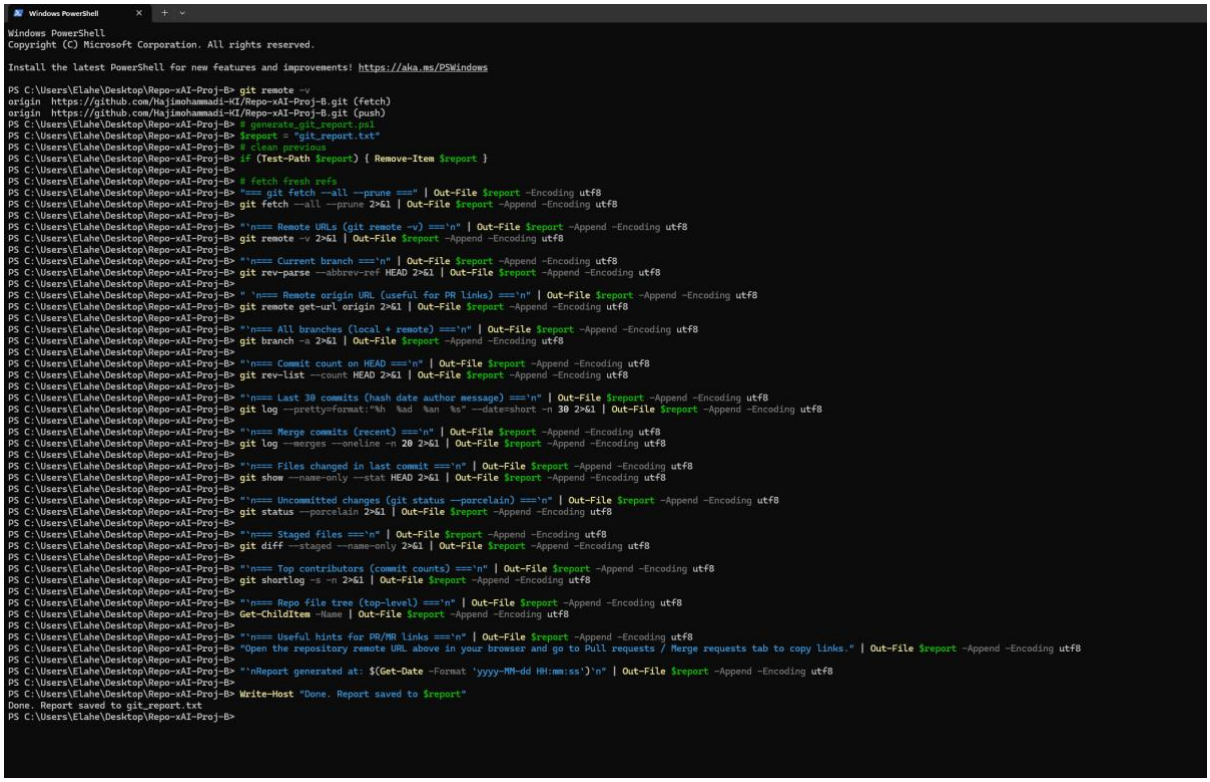


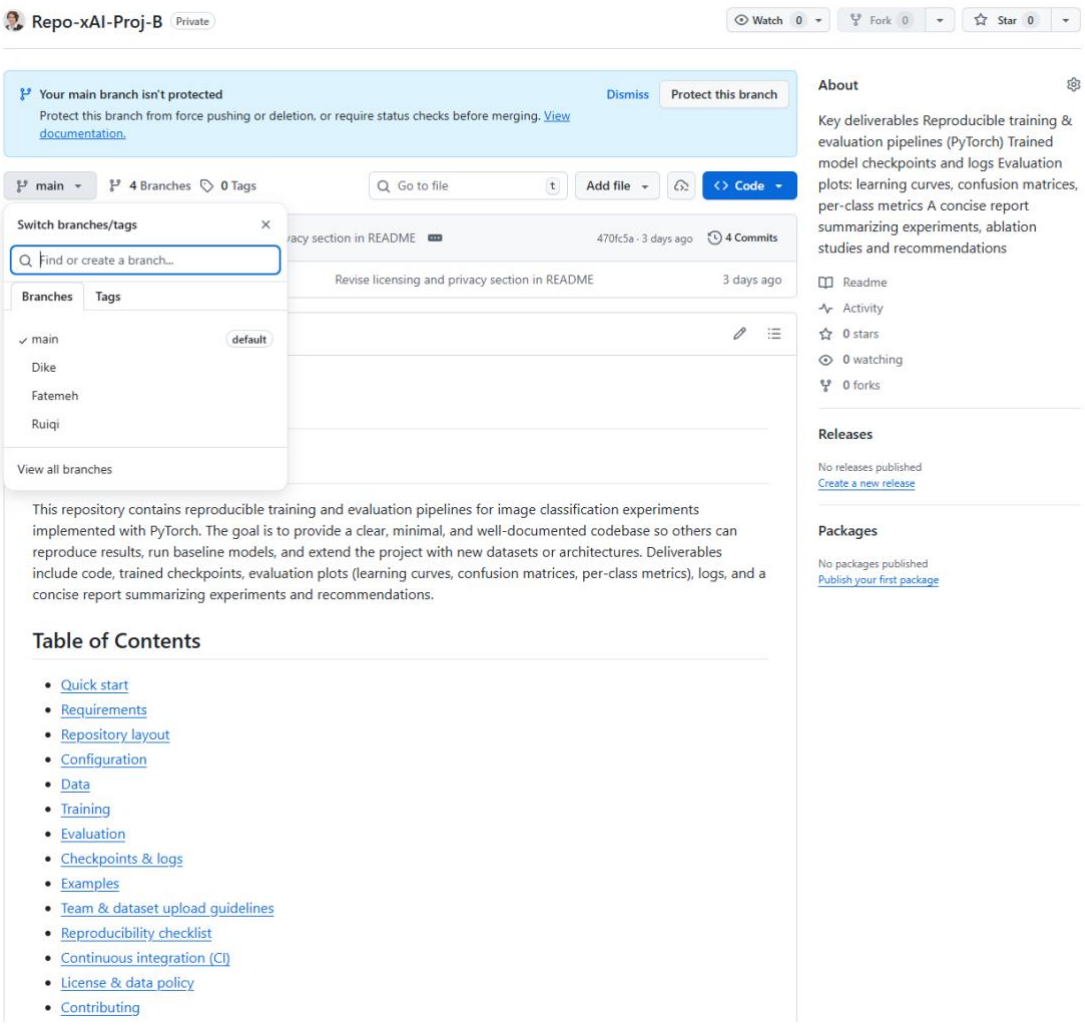
For **project management**, we tested both **Jira** and **Trello**. Based on our actual experience and workflow preferences, we decided to adopt **Trello** as our main project management platform. The current project board is shown below, and a more detailed timeline view can be found in Chapter 4 — *Milestones*

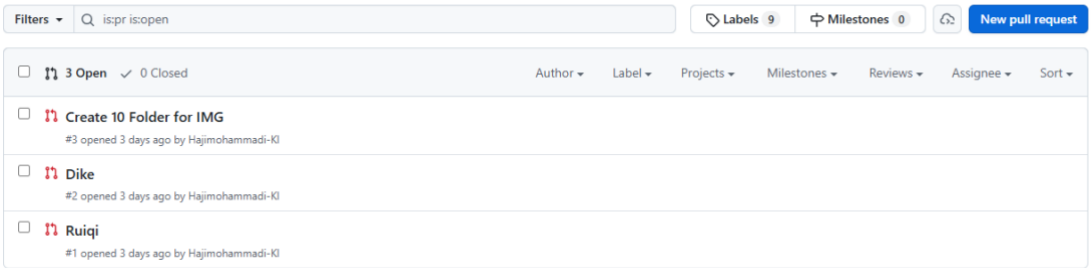
For version control, we compared GitLab and GitHub and ultimately chose GitHub for its higher visibility and accessibility. Following a **feature-branch and merge-request workflow**, each member created an individual branch (e.g., *feat/fatemeh-cnn*, *feat/dike-xxx*, *feat/ruiqi-xxx*) to ensure reproducible changes and maintain a clean code history.

Fatemeh Hajimohammadi set up the shared repository and managed collaborator access, enabling all teammates to push and review code efficiently. She also resolved merge conflicts, wrote clear commit messages, and maintained a consistent repository structure with folders such as *01-data*, *02-models*, *03-training*, *04-eval*, *notebooks*, and *reports*.

We plan to publish our finalized project code on GitHub to enhance transparency and enrich our professional portfolios. The repository homepage is github.com/Hajimohammadi-KI/Repo-xAI-Proj-B, and the branch structure is shown below.







3. Team Roles and Collaboration Plan

Our group consists of three members, each contributing to specific responsibilities. Roles such as Git Master, Scrum Master, and Documentation Master were discussed to maintain efficient workflow and accountability. Even if roles are flexible, we agreed to share knowledge and support one another across all technical and organizational tasks.

Our collaboration plan is initially set to include three weekly progress updates, where each team

member posts their individual progress on Microsoft Teams. In addition, we will hold one team meeting per week, either online or in person, to discuss current tasks and upcoming goals.

Role	Member	Responsibilities
Git Master	*Fatemeh Hajimohammadi*	Creates and maintains the GitLab repository (structure, access rights, .gitignore). Defines a simple branching and merging strategy (main + feature branches). Assists with merge conflicts and reminds team members to commit regularly. Keeps the repository clean and documents it with a short README.
Scrum Master	Ruiqi Sun	Organizes regular team meetings and prepares short agendas. Maintains the Trello / GitLab issue board (Backlog → To Do → Doing → Done). Breaks project requirements into smaller tasks and assigns them. Defines milestones and monitors progress along the timeline. Reports problems or risks (e.g., missing information, scheduling issues).
Communication Master	Dike Gan	Takes meeting notes and uploads them to the shared folder. Keeps documentation consistent (tutorials, tools, role assignment, schedule). Describes how the team collects and manages information from tutorials and literature review. Coordinates communication with the tutor and uploads required files to Team. Ensures that all team members are informed about decisions.

4. Project Timeline and Milestones

We developed a preliminary project timeline with key milestones — including tutorial completion, dataset preparation, baseline model implementation, and model improvement. Setting clear goals helps us monitor progress and adjust our workload. While creating the timeline, we discussed how to balance theoretical study with practical implementation.

Week 1–2: Tool setup & ML fundamentals

- Get familiar with Git/GitHub, Trello, Google Colab, and Python basics.
- Complete the 15 chapters of the Machine Learning introduction course.
- Summarize key ML concepts: datasets, models, training, and evaluation.

Week 3–4: Literature & framework selection

- Review CNN architecture (e.g., ResNet, VGG).
- Decide which framework to use (PyTorch / TensorFlow).
- Discuss and defines the project direction (goal: image classification).

Week 5–6: Baseline model implementation & testing

- Build a baseline image classifier using the shared dataset.
- Train, validate, and test the model to ensure it can **recognize images correctly**.
- Evaluate initial accuracy and note improvement points.

Week 7–9: Model improvement (Including the winter pause)

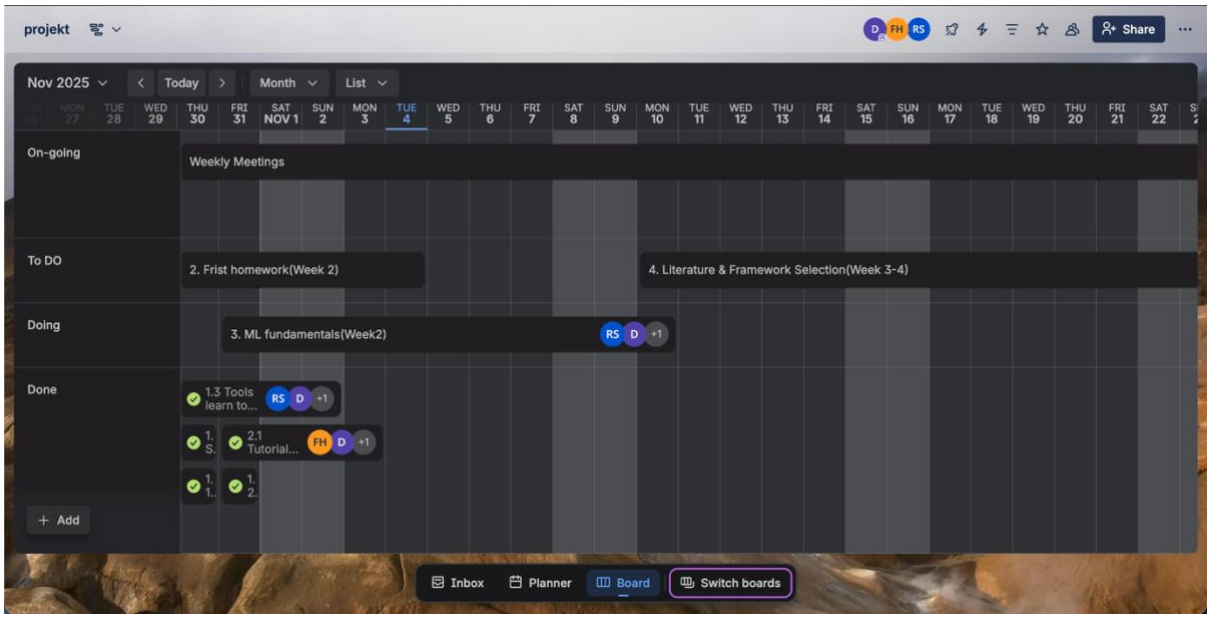
- Apply tuning and data augmentation (e.g., learning rate, flipping, rotation).
- Tune hyperparameters (learning rate, batch size, optimizer)..
- Compare improved results vs. Baseline.

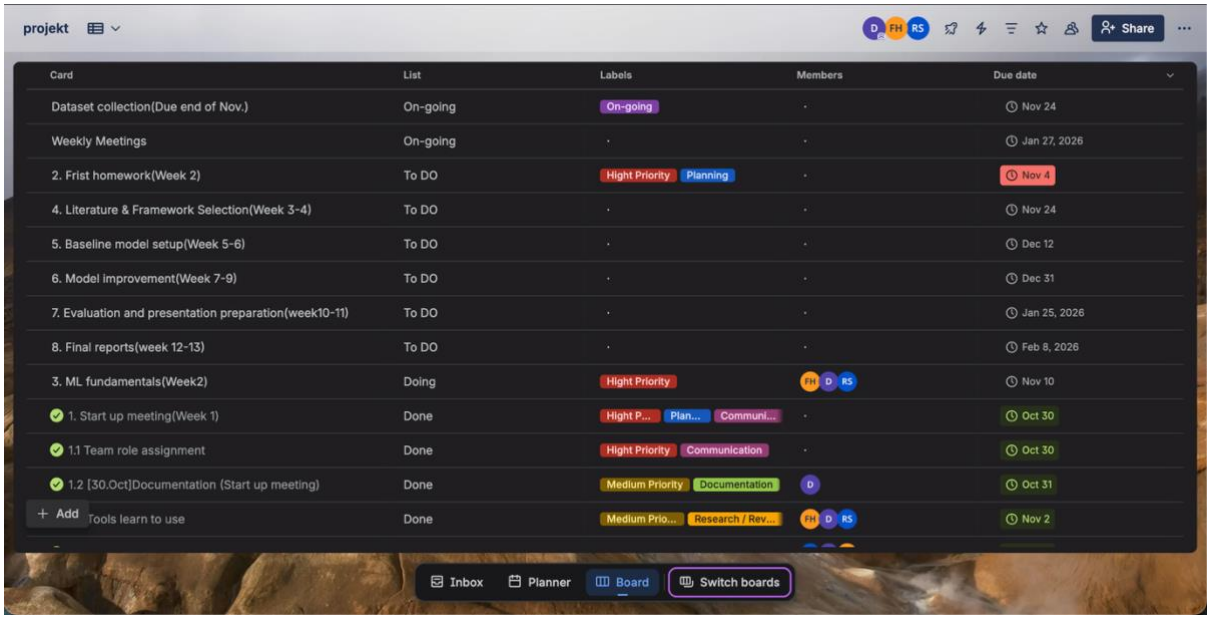
Week 10-11: Evaluation and presentation preparation (week10-11)

- Evaluate final model performance on the full dataset.
- Prepare presentation slides and short technical summary.
- Ensure reproducibility (code + report ready for submission).

Week 12-13: Final reports (week 12-13)

- write own report and submit separately on time


projekt
Share
Nov 2025
Today
Month
List
On-going
Weekly Meetings
To DO
2. Frist homework(Week 2)
4. Literature & Framework Selection(Week 3-4)
Doing
3. ML fundamentals(Week2)
Done
1.3 Tools learn to...
2.1 Tutorial...
Add
Inbox
Planner
Board
Switch boards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Share
Card
List
Labels
Members
Due date
Dataset collection(Due end of Nov.) On-going On-going Nov 24
Weekly Meetings On-going Jan 27, 2026
2. Frist homework(Week 2) To DO Hight Priority Planning Nov 4
4. Literature & Framework Selection(Week 3-4) To DO Nov 24
5. Baseline model setup(Week 5-6) To DO Dec 12
6. Model improvement(Week 7-9) To DO Dec 31
7. Evaluation and presentation preparation(week10-11) To DO Jan 25, 2026
8. Final reports(week 12-13) To DO Feb 8, 2026
3. ML fundamentals(Week2) Doing Hight Priority Nov 10
1. Start up meeting(Week 1) Done Hight P... Plan... Communi... Oct 30
1.1 Team role assignment Done Hight Priority Communication Oct 30
1.2 [30.Oct]Documentation (Start up meeting) Done Medium Priority Documentation Oct 31
Add
Tools learn to use Done Medium Prio... Research / Rev... Nov 2
Inbox
Planner
Board
Switch boards